Project Report – Student Management App Deployment with Docker & Kubernetes

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Project Repository: https://github.com/Payalsingh8277/devops-internship/tree/main/proj-cicd-k8s

1. Introduction

This project demonstrates the deployment of a simple Flask-based Student Management App using Docker and Kubernetes. The project also includes a CI/CD pipeline with GitHub Actions to automate Docker image builds and pushes to Docker Hub. Kubernetes manifests are used to deploy the application locally on Minikube.

2. Abstract

The app allows adding students via a POST request and listing all students via a GET request. Containerization with Docker ensures portability, and Kubernetes handles orchestration and scalability. The CI/CD workflow automates the build and deployment process, providing practical exposure to modern DevOps practices.

3. Tools Used

- **Flask** – Python web framework for the application
- **Docker** – Containerization
- **Kubernetes** – Deployment and service orchestration
- **Minikube** – Local Kubernetes cluster for testing
- **GitHub Actions** – Automated CI/CD pipeline

4. Steps Involved in Building the Project

1. **Flask App Development**
    a. app.py contains routes:
        i. / → Welcome message

ii. `/add` → Add a student (POST JSON data)
iii. `/list` → List all students (GET)

b. `students` stored in memory (list).

2. **Dockerization**
    a. Created a `Dockerfile` to containerize the app
    b. Built image locally and tested with:

```
docker run -p 5000:5000 payalsingh603/proj-cicd-k8s:latest
```

    c. Pushed Docker image to Docker Hub:

```
payalsingh603/proj-cicd-k8s:latest
```

3. **Kubernetes Deployment**
    a. Created deployment manifest `flask-deployment.yaml` with 2 replicas
    b. Created service manifest `flask-service.yaml` of type NodePort
    c. Applied manifests:

```
kubectl apply -f k8s/flask-deployment.yaml
kubectl apply -f k8s/flask-service.yaml
```

4. **CI/CD Automation**
    a. GitHub Actions workflow `cicd.yml` automates:
        i. Docker image build
        ii. Push to Docker Hub on main branch push
5. **Local Testing with Minikube**
    a. Verified deployment with `kubectl get pods`
    b. Accessed service using:

```
minikube service flask-service --url
```